ISCA 2026 · DAC 2026 · ICCAD 2025 · 582 PAPERS

What do this year's chip-design conferences say about where the field is going?

Toward a loop: propose a design, check it with tools, learn, try again.



THE WHOLE STORY

In three sentences. *Each stands alone.*

582

PAPERS BEHIND THREE SENTENCES

Design becomes a loop; accelerators, memory and reliability lead; most evidence is still short descriptions.

1 · THE DIRECTION

Chip design is becoming a lo

Six directions recur across the 582 papers. Most are about feeding checks, measurements and physical limits back into the next design choice.

2 · THE WEIGHT

Accelerators, memory and rel

Largest of 10 themes: architecture and accelerators (103), memory, interconnect and storage (85), power, reliability and security (78).

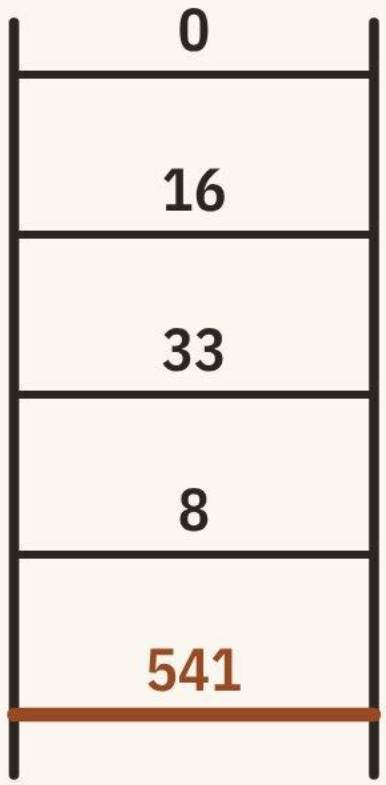
3 · THE CAUTION

Most papers were read from s

541 of 582 records come from program descriptions; 33 from a full paper or preprint. DAC supplies 544 of the 582.

HOW THE ATLAS IS BUILT

From papers to directions, *in five steps.*



Repeated independently
no one has rerun these yet

Checked in depth
a deep-read page each

Full paper or preprint
methods, tables and limits

Abstract, report or released files
the claim, rarely the conditions

Program description only
a short summary from the conference

Each rung up holds fewer papers and says more about each one.

Counted from the repository's paper records

WHERE THE PAPERS COME FROM

DAC is 544 of 582 papers. *The atlas leans toward design tools.*

544

PAPERS FROM DAC 2026

ISCA 2026 adds 17 selected architecture papers. ICCAD 2026 papers were not public yet, so 21 come from ICCAD 2025.

DAC 2026

544

design automation

ICCAD 2025

21

design algorithms

ISCA 2026

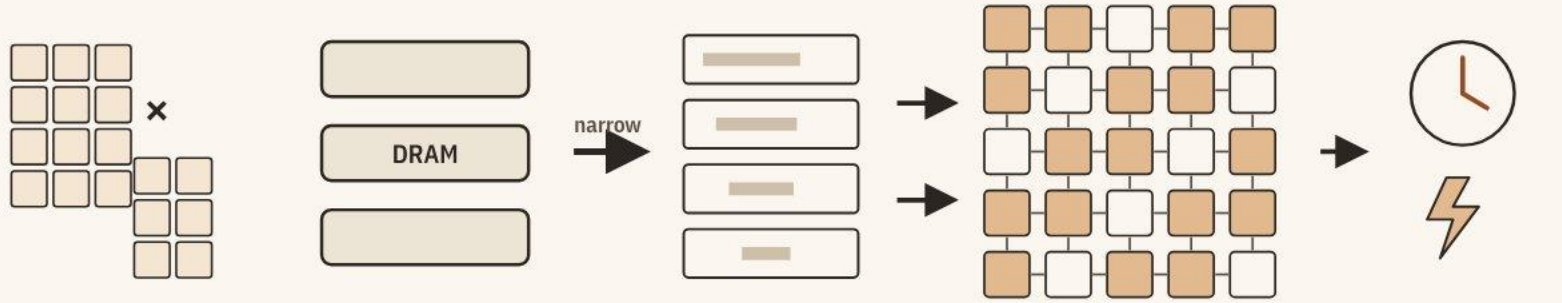
17

computer architecture

Counted from the repository's paper records

WHERE THE ATTENTION GOES

Accelerators, memory and reliability *lead.*



Adding compute units does not help while the narrow pipe stays narrow.

103 PAPERS

Architecture and accelerators

More arithmetic units help only when data arrives in time.

85 PAPERS

Memory, interconnect, and storage

Moving data, not computing it, often sets the time and energy.

78 PAPERS

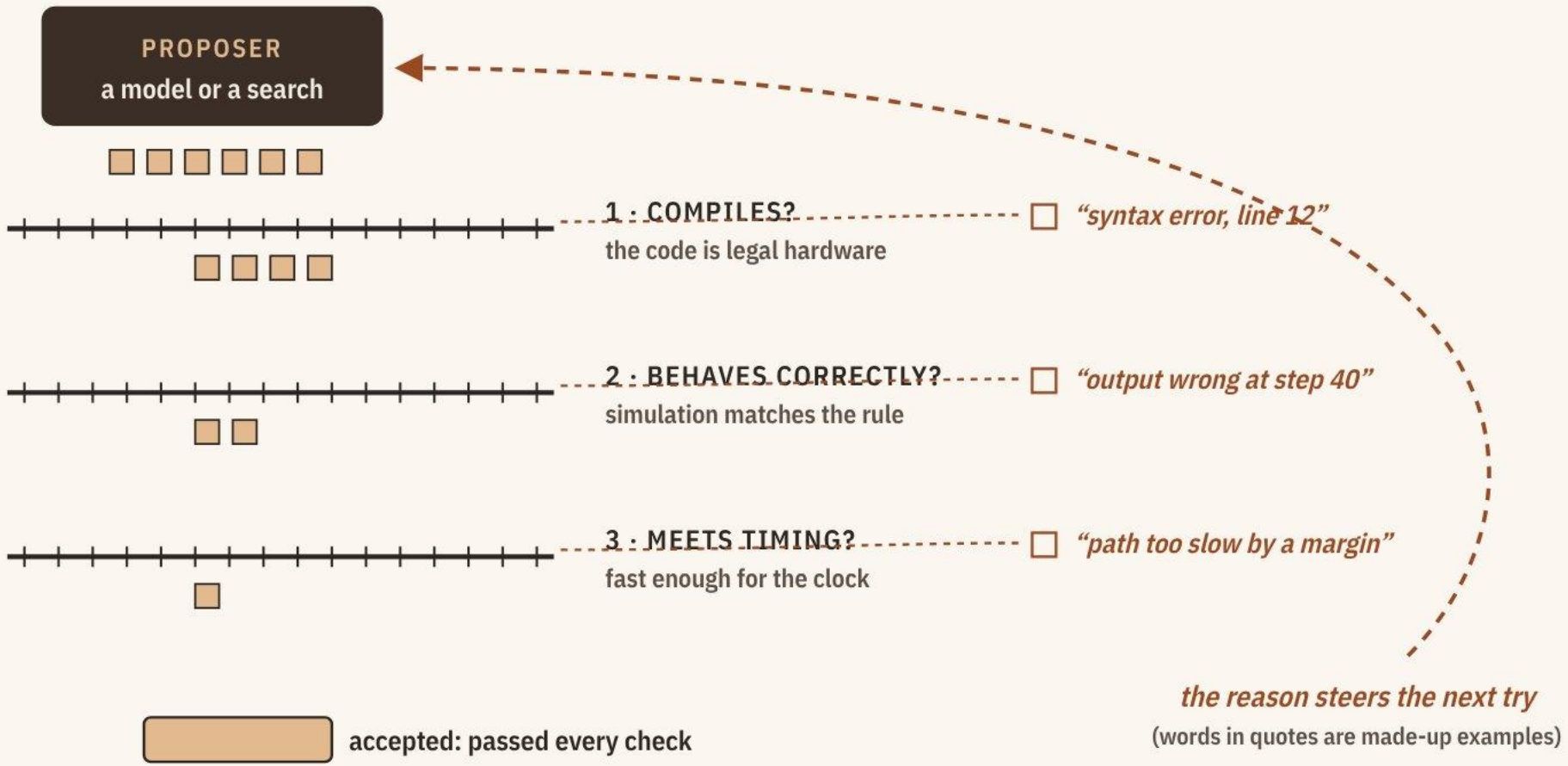
Power, reliability, and security

Protection has a cost, so the fault or attack must be named first.

Counted from the repository's paper records

THE COMMON THREAD

Design becomes a loop. *Checks steer the next try.*

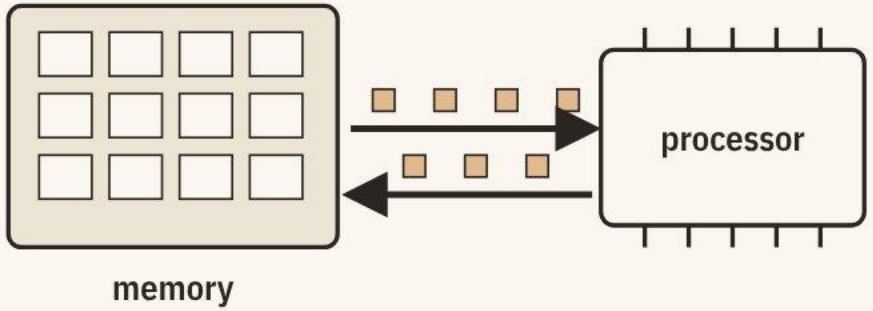


Our drawing; the quoted messages are made-up examples

DATA MOVEMENT

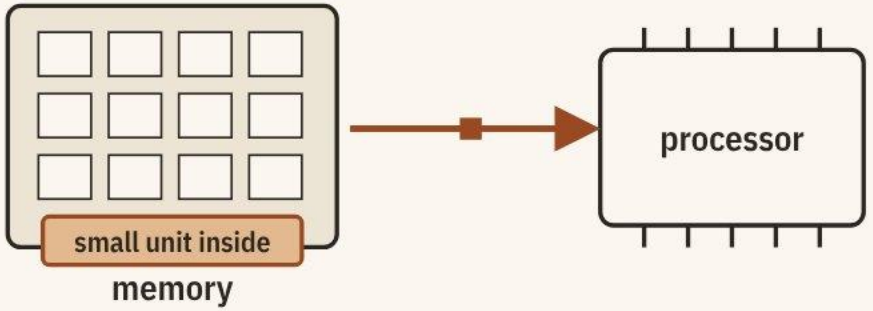
Moving data is often *the real cost.*

SEND THE DATA TO THE WORK



many numbers make the long trip, twice

SEND THE WORK TO THE DATA



only the short result travels

The test: time and energy for the whole job, with every copy counted.

PHYSICS AND PROTECTION

Heat, wires, faults and attacks *move earlier.*

PHYSICAL REALITY

Heat, wires, power

1. A fast estimate guides early choices.
2. An accurate check confirms the final design.

The fast model must agree with the slow one where it matters.

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s

FAULTS AND ATTACKS

Named first

1. The threat or fault is stated in the requirement.
2. Its cost is shown with protection on.

Useful correct work, not raw speed.

THE TEST **Does the early estimate still hold after the full, accurate check?**

Our reading of directions 4 and 5

MEASUREMENT

Better ways *to measure results.*

TASK	RULE FOR SUCCESS	RESULT, WITH SPREAD	REPEATED BY OTHERS?
design A	passes all tests (v2)		☐
design B	passes all tests (v2)		☐
design C	meets timing (v2)		☐
design D	meets timing (v2)		☐

A score means little without its tasks, its rule, its version and its spread.

(task names and dots are made-up examples)

HOW SURE WE ARE

541 of 582 records *come from short descriptions.*

33

PAPERS READ FROM A FULL PAPER OR PREPRINT

A program description gives the problem and the claim, rarely the test conditions. 16 papers were checked in depth. None was repeated independently.

DESCRIPTIONS	ABSTRACTS, REPORTS	FULL PAPERS	DEEP READS
541	8	33	16
short program text	the claim	methods and limits	a page each

Counted from the repository's paper records

WHAT TO WATCH

Six directions, *each with its disproof.*

6

HYPOTHESES, NONE PROVEN

Each direction page names its evidence, the paper that pushes back, and the result that would disprove it.

DIRECTION 1

Requirements checked while designing

DIRECTION 2

Moving data is the real cost

DIRECTION 3

Measure, then decide the next design

DIRECTION 4

Plan for faults and attacks early

DIRECTION 5

Heat, wires and power in the loop

YOUR TURN

Start with a theme, *then follow its papers.*

Every page explains the problem before the papers, and says how deep its sources go.

HTTPS://MEHTAMA1234.GITHUB.IO/SEMICONDUCTOR-CONFERENCES/